

Adam O'Deven

Mechanical Engineering & Industrial Design student

focused on translating user needs into manufacturable systems through CAD and prototyping. Experienced in owning early-stage hardware from concept through physical validation.

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(248) 829-0272

Skills

CAD & Mechanical Design

- SolidWorks (7 years)
- Parametric modeling
- Top-down assembly design
- DFM & DFA
- Familiar with GD&T

Prototyping & Fabrication

- CNC machining (3-axis)
- 3D printing (FDM)
- Manual Milling
- Metal lathe
- Mechanical assembly
- Soldering
- Familiar with circuit design

Design & Visualization

- Concept sketching
- Adobe Suite
- KeyShot
- Product photography

Research & Evaluation

- User research
- Usability testing
- Prototype evaluation
- Iterative refinement

Education

Georgia Institute of Technology – June 2024 - Present

- BS Mechanical Engineering – BS Industrial Design – 3.75 GPA
- Honors Program – GT Crew – Exec Board @ Chi Phi Fraternity

Notable Coursework: Statics, Dynamics, User-Centered Design Methods, Circuits & Electronics, Computing for Engineers, Intro to Eng. Graphics, Statistics & Applications.

Experience

REAR Lab Research – Aug 2024 - Dec 2024

Product Developer

- Informed early-stage assistive medical device prototypes by translating clinical user observations into mechanical requirements and CAD assemblies.
- Built and iterated multi-part assemblies in SolidWorks and Fusion 360, incorporating ergonomic constraints, component packaging, and rapid prototyping limitations.
- Fabricated and assembled prototypes using 3D printing, manual machining, and mechanical assembly.
- Conducted field observation and usability testing with clinicians and wheelchair users at the Shepherd Center to validate mechanical feasibility and user interaction.
- Documented and compared CAD workflows to reduce iteration time and improve design handoff between conceptual and detailed modeling stages.

RoboJackets K12 Foundation – June 2018 - Present

Mentor, Mechanical Engineer

- Led mechanical design, prototyping, and assembly for 28 competition systems, balancing robustness, manufacturability, and performance under time and resource constraints.
- Owned assembly development from SolidWorks to implementation, emphasizing parametric modeling, assemblies, and design intent for rapid iteration and repairability.
- Supported system production across CNC machining, 3D printing, manual machining, and electrical/mechanical assembly.
- Coordinated build and assembly work across a 25-person interdisciplinary team.

Georgia Tech Office of IT – Aug 2024 - Present

Wireless Network Services Student Assistant

- Supported campus hardware deployments by documenting technical processes and troubleshooting physical networking equipment.